



AFRC NEWS

Air Force validates open architecture, expands Collaborative Combat Aircraft ecosystem



Published Feb. 12, 2026

Secretary of the Air Force Public Affairs

ARLINGTON, Va. (AFNS) -- The Air Force continues accelerating its Collaborative Combat Aircraft program by implementing the government-owned Autonomy Government Reference Architecture across multiple vendor platforms. This effort validates a cornerstone of the service's acquisition transformation: using a modular, open-systems approach to supercharge the defense industrial base.



[PHOTO DETAILS](#) / [DOWNLOAD HI-RES](#)

This initiative is a prime example of the department's new approach to acquisition — one that prioritizes speed, innovation and a software-first mindset. The A-GRA is being integrated by the program's mission autonomy vendors, RTX Collins and Shield AI, which have started semi-autonomous flight testing in partnership with General Atomics on the YFQ-42 platform and Anduril on the YFQ-44, respectively. By proving the architecture functions effectively across different airframes and mission autonomy from different vendors, the Air Force is demonstrating that mission software can be decoupled from specific vehicle hardware, breaking down barriers for technology integration and

fostering a more competitive and innovative ecosystem.

"Verifying A-GRA across multiple partners is critical to our acquisition strategy," said Col. Timothy Helfrich, Portfolio Acquisition executive for Fighters and Advanced Aircraft. "It proves that we are not locked into a single solution or a single vendor. We are instead building a competitive ecosystem where the best algorithms can be deployed rapidly to the warfighter on any A-GRA compliant platform, regardless of the vendor providing the algorithm.

The A-GRA is a Modular Open System Approach, designed to prevent "vendor lock" by establishing a universal standard for mission autonomy. This allows the Air Force to rapidly onboard new software and algorithms from a diverse range of traditional and non-traditional industry partners. This approach is central to the Air Force's effort to create a more agile and lethal force, capable of outpacing evolving threats.

"We are seeing the vision of a modular, adaptable force come to life," the Agile Development Office director said.

"Integrating A-GRA onto multiple platforms so

quickly demonstrates that our open-system approach works. It allows us to iterate tactics and capabilities across the fleet at a pace that keeps us ahead of the threat.”

As the program advances, the continuous evaluation of the A-
GRA provides the foundational data needed to field a decisive
operational capability. This effort directly supports
the **National Defense Strategy** by developing platforms that
complement major weapons systems and ensures the Air
Force remains on track to deliver an affordable, adaptable,
and potent force multiplier for the joint force.



[PHOTO DETAILS](#) / [DOWNLOAD HI-RES](#)



[PHOTO DETAILS](#) / [DOWNLOAD HI-RES](#)

USAF. (U.S. Air Force Graphic by Rosario "Charo" Gutierrez)

